The Boundary-Balance Hinge

Inflow, outflow and accumulation in three physical systems

Name: ____________________

Class: ____________________

IB DP Physics: thermal physics, climate balance and circuits

Working rule:

input rate − output rate = rate of accumulation.

Choose the boundary first. At steady state the accumulation term is zero. Within a combined isolated system, transfers between its parts cancel in the total ledger.

1. Planet and atmosphere: radiation across a boundary

The solar irradiance at Earth is $S = 1360\,\mathrm{W\,m^{-2}}$. Take the planetary albedo as $\alpha = 0.30$ and use $\sigma = 5.67 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$. Treat Earth as a sphere in radiative equilibrium.

A. Choose the system boundary

Draw a boundary around Earth and its atmosphere. Show incoming solar power, reflected solar power, outgoing infrared radiation, and atmospheric infrared emitted upward and downward. Which arrows cross the chosen boundary?

B. Build the steady-state ledger

Explain why Earth intercepts sunlight over area πR^2 but emits over area $4\pi R^2$. Write

$$(1-\alpha)S\pi R^2 = 4\pi R^2 \sigma T_e^4$$

and calculate the effective emission temperature T_e. Show that R cancels.

C. Locate atmospheric re-radiation

The atmosphere absorbs and re-emits infrared radiation. Explain why upward and downward radiation exchanged between the surface and atmosphere is internal to the chosen system. Why can the surface temperature differ from T_e while the top-of-atmosphere balance still holds?

D. Interrogate equilibrium

If absorbed solar power temporarily exceeds outgoing infrared power, state the sign of the accumulation term and predict the direction of temperature change.

2. Parallel circuit: charge at a junction

A 12.0 V supply is connected to two parallel resistors, $R_1 = 6.0\,\Omega$ and $R_2 = 4.0\,\Omega$. The circuit is in steady state.

A. Draw the junction boundary

Sketch the two branches and draw a small boundary around the splitting junction. Label I_{in}, I_1 and I_2. State what would happen physically if charge entered the boundary faster than it left.

B. Supply the branch model

The potential difference across each parallel branch is 12.0 V. Use $I = V/R$ to calculate I_1 and I_2. Then apply charge conservation:

$$\frac{\Delta Q_j}{\Delta t} = I_{\text{in}} - I_1 - I_2.$$

Find I_{in} for steady state.

C. Follow charge over an interval

During 0.50 s, calculate the charge entering the junction and the charge leaving through each branch. Verify the charge ledger numerically. Explain why current is a rate of charge transfer rather than an amount of charge stored in a branch.

D. Separate rule from closure

Which result follows from charge conservation? Which results require Ohm's law and the fact that parallel branches share the same potential difference?

3. Thermal contact: two objects become one system

A 0.200 kg sample of water at $T_1 = 20.0^\circ\mathrm{C}$ is placed in an insulated container. A 0.100 kg aluminium block at $T_2 = 100.0^\circ\mathrm{C}$ is added. Use $c_w = 4200\,\mathrm{J\,kg^{-1}\,K^{-1}}$ and $c_{\mathrm{Al}} = 900\,\mathrm{J\,kg^{-1}\,K^{-1}}$.

A. Choose two boundaries, then combine them

Draw separate boundaries around the water and block, with an energy-transfer arrow between them. Then draw one boundary around both. Identify which transfer becomes internal when the two objects are treated as one system.

B. Write the combined energy ledger

For the insulated combined system,

$$Q_w + Q_{\mathrm{Al}} = 0, \qquad Q = mc\Delta T.$$

Use the common final temperature T_f to write one equation and calculate T_f. Keep the signs of both temperature changes explicit.

C. Verify the internal transfer

Calculate the energy gained by the water and lost by the aluminium. Compare their magnitudes. Explain why thermal energy is conserved for the combined system while the temperature of each object changes.

D. Interrogate the assumptions

List three assumptions needed for the calculated T_f: insulation, constant specific heat capacities, negligible container heat capacity, or complete thermal equilibrium.

Synthesis: conservation becomes operational when the boundary and transfer law are explicit

1. Complete the comparison.

Case	Quantity balanced	Boundary condition	Relation supplying the terms	Accumulation zero means
Earth–atmosphere				
Circuit junction				
Water + block				

2. Develop the common structure.

Use all three cases to explain:

"The boundary decides which transfers appear in the ledger."

Include the terms *input*, *output*, *accumulation*, *steady state*, *internal transfer*, *junction*, *equilibrium*, and *closure*. State what conservation determines and what the case-specific equations supply.

Teacher guide and indicative answers

The Boundary-Balance Hinge

1. Planet and atmosphere

Boundary. With Earth and atmosphere inside the boundary, absorbed solar radiation enters and outgoing infrared radiation leaves. Reflection also leaves. Surface infrared, atmospheric absorption and downward atmospheric radiation are internal exchanges.
Balance.

$$(1-\alpha)S\pi R^2 = 4\pi R^2\sigma T_e^4$$

so

$$T_e = \left[\frac{(1-\alpha)S}{4\sigma}\right]^{1/4} \approx 255\,\mathrm{K}.$$

The factor of four follows from intercepting over a disc and emitting over a sphere.
Atmosphere. The effective emission temperature is associated with radiation escaping to space. Greenhouse gases redistribute energy and alter the altitude from which infrared escapes. The surface can therefore be warmer while the top-of-atmosphere steady-state ledger remains absorbed solar = outgoing infrared.
Accumulation. If input exceeds output, stored internal energy increases and the system warms. Rising temperature tends to increase thermal emission through the Stefan–Boltzmann relation.
Hinge point. Conservation requires equality at equilibrium. Albedo, geometry and the radiation law determine the numerical terms.

2. Parallel-circuit junction

Each branch has the supply potential difference:

$$I_1 = \frac{12.0}{6.0} = 2.0\,\mathrm{A}, \qquad I_2 = \frac{12.0}{4.0} = 3.0\,\mathrm{A}.$$

At steady state there is no continuing charge accumulation at the junction:

$$0 = I_{\text{in}} - I_1 - I_2, \qquad I_{\text{in}} = 5.0\,\mathrm{A}.$$

During 0.50 s,

$$Q_{\text{in}} = I_{\text{in}}t = 2.50\,\mathrm{C},$$

$$Q_1 = 1.00\,\mathrm{C}, \qquad Q_2 = 1.50\,\mathrm{C},$$

and $Q_{\text{in}} = Q_1 + Q_2$.
If inflow exceeded outflow, charge would briefly accumulate, changing the local electric field and potentials until the currents adjusted. In an ordinary steady circuit this transient is extremely short.
Hinge point. Charge conservation supplies the junction equation. Ohm's law and the shared branch voltage supply the branch currents. Current represents charge crossing a boundary per unit time.

3. Thermal contact

For the combined insulated system, energy transferred out of the aluminium is transferred into the water:

$$m_w c_w(T_f - T_1) + m_{\text{Al}}c_{\text{Al}}(T_f - T_2) = 0.$$

Substitution gives

$$0.200(4200)(T_f - 20.0) + 0.100(900)(T_f - 100.0) = 0,$$

so

$$T_f \approx 27.7^\circ\mathrm{C}.$$

Water gain:

$$Q_w = 0.200(4200)(27.7 - 20.0) \approx 6.50\times10^3\,\mathrm{J}.$$

Aluminium change:

$$Q_{\text{Al}} = 0.100(900)(27.7 - 100.0) \approx -6.50\times10^3\,\mathrm{J}.$$

Temperature is a state variable for each object and need not be conserved. The combined thermal-energy ledger closes because the exchange is internal and external transfer is assumed negligible.
Assumptions. Insulation; constant c values; negligible container heat capacity; no phase change; complete thermal equilibrium.
Hinge point. Conservation gives $\sum Q = 0$. The relation $Q = mc\Delta T$ and the common final temperature supply the closure.

Teaching focus and synthesis

Completed comparison

Case	Quantity	Zero accumulation	Closure
Earth–atmosphere	Energy	absorbed solar power = outgoing infrared power	geometry, albedo and $P = \sigma AT^4$
Junction	Charge	$I_{\text{in}} = I_1 + I_2$	shared voltage and $I = V/R$
Water + block	Energy	gain by water + loss by block = 0	$Q = mc\Delta T$ and common T_f

Expected synthesis. Students should identify one balance architecture with three meanings. The planetary case is a steady energy-flow balance across an external boundary. The junction case is a charge-continuity balance at a node. The thermal case combines two objects so that their mutual transfer becomes internal. In each case, conservation constrains the ledger while a specific physical relation calculates its terms.
Useful diagnostic questions. What is inside the boundary? Which arrows cross it? Can the conserved quantity accumulate? Is the system steady, isolated or approaching equilibrium? Which relation converts the physical description into a number?
Common misconception. Equal input and output describe zero accumulation, not an absence of transfer. At equilibrium or steady state, substantial two-way or through-flow transfers may continue while the stored total remains constant.